Final Project Proposal

Diversified E-Commerce Dataset Analysis

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Introduction and Project Goals

The chosen dataset provides a comprehensive overview of an online e-commerce platform, including details about orders, customers, products, pricing, shipping, and payment methods. With nearly one million transaction records, it presents an excellent opportunity to uncover meaningful insights about consumer behavior, product performance, and operational efficiency.

The main goal of this visualization project is to help the audience understand how various business dimensions—time, product categories, customers, payments, and shipping—contribute to overall performance. Through this analysis, the dashboard will reveal key sales trends, customer purchase patterns, profitability insights, and satisfaction drivers. The resulting visuals will provide valuable business intelligence that could guide decision-making in pricing, inventory management, and marketing strategy.

Description of the Visualization Problem and Dataset

The visualization challenge centers on exploring how e-commerce activities vary across different dimensions such as time, geography, product categories, and customer segments. The dataset contains around 1,000,000 individual records, where each record represents a unique order placed by a customer.

Each record includes key attributes such as order details (Order_ID, Order_Date), product information (Product_ID, Category, Price, Quantity), customer details (Customer_ID), payment and shipping choices, and performance indicators like review ratings.

In Tableau, multiple calculated fields will be created to enhance insights — such as Total Sales, Profit, Profit Margin, Average Order Value (AOV),

Customer Segments, and Weekday vs. Weekend performance. These will allow for more sophisticated measures beyond raw data.

Dataset Structure and Field Overview

Field Name	Type	Description	Data Type
Order_ID	Discrete	Unique identifier for each order	String
Customer_ID	Discrete	Unique customer identifier	String
Product_ID	Discrete	Unique product identifier	String
Category	Discrete	Product category name	String
Price	Continuous	Price per product	Float
Quantity	Continuous	Quantity ordered	Integer
Payment_Method	Discrete	Payment mode used	String
Shipping_Type	Discrete	Type of shipping selected	String
Review_Rating	Continuous	Customer rating (1–5 scale)	Float
Order_Date	Continuous (Date)	Date when order was placed	Date-Time

Exploratory Questions:

Time-Based Analytics

1. How have total sales and total orders evolved over time (month/year)?
2. Which months or quarters receive the highest average review ratings?
3. How do sales differ between weekdays and weekends?

Product Performance Analytics

4. Which product categories contribute most to total sales and profit margins?
5. What is the relationship between product price and review ratings?
6. Which categories attract the most repeat purchases?

Customer Behavior & Segmentation

7. Who are the top 10 customers based on total spend?
8. What is the Average Order Value (AOV) across customer segments?

Shipping & Logistics Analytics

9. How do total sales and average ratings differ across shipping types?
10. How does delivery duration (if derived) affect customer ratings?

Payment & Transaction Insights

11. Which payment methods generate the highest total revenue?
12. What is the average transaction value per payment type?

Profitability & Business Insights

13. Which product categories are most and least profitable?
14. How do profit margins vary across shipping types and payment methods?

Geographic Insights

15. Which regions show the highest sales and average customer ratings?
16. How does order volume vary regionally over time?

Expected Outcome

By the end of this project, the visualization will present a multi-faceted dashboard that allows the user to:

- Identify top-performing categories, customers, and payment types.
- Analyze sales trends and seasonal performance.
- Evaluate customer satisfaction and operational efficiency.
- Understand profit distribution through calculated metrics and dual-axis visuals.

This proposal demonstrates a structured, analytical approach that combines data storytelling, advanced Tableau skills, and business intelligence thinking—ultimately transforming raw data into actionable insight.